

Micro I, class 11

We briefly reviewed properties of the cost function that we already knew from our discussion of consumer theory: for each $q \geq 0$, it is nondecreasing, homogenous of degree 1, and concave. We then considered the shape of the cost function as a function of *output* for simple, single-output, single-input technologies. The conclusions here are that *nondecreasing returns to scale* implies that average cost ($c(q, w)/q$) is nondecreasing in output; that *nondincreasing returns to scale* implies that average cost ($c(q, w)/q$) is nonincreasing in output; and that a *concave* production function implies that cost is *convex* in output (marginal cost is increasing). These facts generalize to technologies with many inputs, and I leave some of those to you in ASMT Va (in particular, that, if the technology exhibits constant returns to scale, then the cost function is linear; and that, if the production function is concave, then the cost function is convex).

I then introduced the two-good quasilinear economy, our last topic before the midterm. Next week we will take up the (Pareto) optimality and existence of a competitive equilibrium. Existence will follow from an application of the *intermediate value theorem* from elementary calculus, a special kind of a fixed-point theorem.

Details are in my notes on the two-good quasilinear economy.